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Contrast and brightness balance in image enhancement using Cuckoo Search-optimized image fusion

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ABSTRACT

Many vision-based systems suffer from poor levels of contrast and brightness, mainly because of inadequate and improper illumination during the image acquisition process. As a result, the required specified information from the acquired image is not available for the particular application. In general, it is hard to achieve a balance between the improvement of contrast and brightness in image enhancement. By introducing nature-inspired optimization in image enhancement, the best features of the image are utilized, and the complexity related to the nonlinearity of images can be solved with various constraints, like a balance between contrast and brightness. In this work, a novel automatic method for image enhancement to find a balance between contrast and brightness is developed by using Cuckoo Search-optimized image fusion. First, the Cuckoo Search-based optimization algorithm generates two sets of optimized parameters. These parameter sets are used to generate a pair of enhanced images, one with a high degree of sharpness and contrast, the other is bright and has been improved without losing the level of detail. Furthermore, the two enhanced images are fused by the fusion process to obtain an output image where the contrast and brightness are in balance. The effectiveness of the proposed method is verified by applying it to standard images (CVG-UGR image database) and lathe tool images. Experimental results demonstrated that the proposed method performs better with regard to both the quality of contrast and brightness, moreover, yields enhanced quality evaluation metrics compared to the other conventional techniques.

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1. Introduction

Image enhancement (IE) is a vital tool in image processing because it enhances the quality of images and draws attention to image features. Additionally, it is mostly used as a primary processing tool across multiple applications and machine vision systems. Image enhancement is a process in which the pixel intensity value of the input image is transformed to the new intensity value so that it highlights certain features for subsequent image analysis or display. When an enhancement is performed for display reasons, it increases the contrast and sharpness of the image, emphasizing its visual appeal. On the other hand, when the enhancement is performed for the purpose of machine perception, as object identification by machine, the enhancement emphasizes certain features, e.g., edges, boundaries, and textures, so that the image analysis system measures quantitative information from enhanced images and uses this to make a sophisticated decision for a particular application. In general, image enhancement methods can be broadly divided into two categories: Spatial domain methods and Frequency domain methods (Gonzalez et al., 2011). Spatial domain methods directly operate on the grayscale value of an input image. In frequency domain methods, the input image is first transformed by applying the Fourier transform, which is then modified by a predefined operation, and the output image is obtained by computing the inverse of the transform domain. Histogram equalisation (HE), linear contrast stretching (LCS), and unsharp masking (UM) are the three most commonly used techniques for image enhancement. They are simple to apply in image processing.

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HE improves the global contrast by distributing the most frequently occurring intensity values throughout the entire image. Linear contrast stretching is a process of increasing the dynamic range of an image's intensity level. In many cases, because these methods deal with the image on a global scale, the local information is not extracted. Poor and improper illumination during the image acquisition process causes poor levels of contrast and brightness. Contrast is defined as a separation in luminance (intensity level) between the dark and bright areas of an image. An improvement in contrast increases this separation so that the object becomes distinguishable (Agrawal et al., 2022). Brightness is defined as the specified amount of luminance in an image. Both enhancements are affected by each other. In general, it is difficult to balance between these improvements in image enhancement.

In this paper, a novel automatic method of image enhancement is developed by using a Cuckoo Search-based algorithm. A new parametric transformation function has been introduced to enhance the details of overexposed and underexposed images. The contribution of this paper is bulleted as follow:

- The aim is to enhance the poor exposure and poor contrast of the image by improving the brightness as well as enhancing the changes between light and dark by increasing or decreasing the contrast.
- The Cuckoo Search algorithm is used to determine the optimal set of parameters without a human interpreter.
- The proposed method generates two images - one rich in details like high entropy, a large number of edge pixels and high degree of sharpness; the brightness of the other has been adjusted. A fusion process further combines these two images to generate an enhanced image that is visually pleasing with a better balance between brightness and contrast.

The Cuckoo Search (CS) algorithm has been widely used to solve various engineering optimization problems. It is simple, robust, scalable, flexible, and speedy in nature. Moreover, the CS algorithm features an appropriate balance of exploration and exploitation, ensuring the algorithm's optimal performance. Various modifications can be found in the literature related to strategy, rate, and control over the exploration and exploitation of the algorithm during the search process (Kordestani et al., 2018; Sait et al., 2016). Peng, 2021 introduced a multi-strategy serial cuckoo search (MSSCS) by the inspiration of serial behaviors in cuckoo growth. Also, a new variant was proposed by combining Spark framework (He, 2019). Here, a simple framework of Cuckoo Search is designed to search for two nests that have different qualities. The quality of a nest is defined by two different objective functions. These two objective functions utilize the contrast and brightness of an image for image enhancement. The results are good in both quality and quantity.

2. Related work

In machine vision, the quality of the captured image can also suffer as a result of improper illumination. It may be underexposed (dark and dull) or overexposed (bright and saturated). Even under the best environmental conditions with optimal camera settings, it is practically impossible to exactly match the captured image to the observed scene. It is possible that exposure artifacts are introduced which may wash out the details of the image. A comprehensive details of image enhancement techniques is mentioned by (Qi, et al., 2021). In the literature, researchers have shown an interest in developing an image enhancement method concerning exposure problems using contrast adjustments and gamma correction. (Vidya et al., 2012) proposed a selective illumination enhancement technique (SIET) for normalizing light variation throughout an image. (Huang et al., 2013) presented a method, termed as AGCWD, to enhance the brightness of the low illumination image by combining histogram analysis, weighting distribution, and gamma correction.

(Singh and Kapoor, 2014) presented a concept of an exposure threshold into the histogram equalization to enhance the low-exposure grayscale image, termed as ESIHE. (Arriaga-Garcia et al., 2014) presented a simple method, called as BEASF, for enhancing contrast and preserving brightness by dividing the histogram into sub-histograms and smoothing their cumulative distributions using the sigmoid function. G. Jiang et al. also explain that contrast and brightness are two important but conflicting objectives. They proposed a balanced performance in terms of brightness and contrast enhancement by adopting nonlinear gamma correction and weighted-sum approach, which is termed as CEGAMMA (Jiang et al., 2015). Turgay Celik proposed a new global enhancement technique named Spatial Entropy-based Contrast Enhancement (SECE) (Celik, 2014) without any parameter selection. Furthermore, Turgay also implemented a method called SECEDCT (Celik, 2014) that uses SECE and the 2D discrete cosine transform to achieve both global and local enhancement. (Guo et al., 2017) proposed an enhancement technique based on estimating the illumination map for images suffer from poor lighting and called LIME. (Ying et al., 1711) proposed a fusion framework to fuse the images with multiple exposure for the enhancement of low-light images and referred to as BIMEF. In recent years, in the area of IE, deep learning approaches have also emerged. (Shen, et al., 1711) proposed an enhancement model for the low-light images by implementing the multi-scale Retinex through a convolutional neural network (CNN). (Wei, et al., 1808) designed a Retinex deep network made of Decomposition network and Enhance network for enhancement in images suffered by poor lighting.

Recent advancements in image enhancement have included the use of nature-inspired metaheuristic algorithms to automate it by optimizing the parameters of the algorithm by some defined objective function. (Pal et al., 1994) proposed a Genetic Algorithm (GA)-based image enhancement with the combination of different transformation functions. (Hashemi, 2010) applied an IE technique using a simple chromosome structure and genetic operators. (Munteanu and Rosa, 2004) proposed an automatic process for the enhancement of grayscale images driven by an evolutionary algorithm. Similarly, several metaheuristic algorithms, e.g. Particle Swarm Optimization (PSO) (Gorai and Ghosh, 2009), Firefly Algorithm (FA) (Hassanzadeh et al., 2011), Differential Evolution (DE) (Sarangi, 2014), Harmony Search (Al-Betar et al., 2016), etc., have been successfully applied to image enhancement. In these methods, a parametric local enhancement kernel is used, which results in a high degree of variation in the features of the input images. One of the recent nature-inspired algorithms widely used in various applications is the Cuckoo Search (CS) algorithm developed by (Yang and Suash, 2009). Studies show that it performs far better than PSO and GA. (Civicioglu and Besdok, 2013) compared the CS algorithm performance and success rate with PSO, DE and ABC, statistically. The findings were in favor of CS and DE compared to the PSO and ABC. Additionally, the successful implementation of CS in the various applications was reviewed by (Yang and Deb, 2014). Moreover, it outperforms several other algorithms in various studies (Abdualrhman and Padma, 2019; Garg and Sahu, 2015; Samal et al., 2022; Agrawal et al., 2012). used the CS with a parametric local enhancement kernel for image enhancement. (Babu and Sunitha, 2015) proposed a CS based contrast enhancement followed by morphological processing to emphasize the image details. CS was also employed to improve the image quality by (Ghosh et al., 2014). (Bhandari et al., 2014) proposed a contrast enhancement for low-resolution satellite images by using CS

algorithm to optimize the parameters in the adaptive thresholding function in DWT-SVD technique. Further, (Suresh et al., 2017) proposed a novel adaptive cuckoo search to enhance the contrast in the satellite image. (Kamoona and Patra, 2019) utilized the greedy selection approach in cuckoo search algorithm for gray scale image enhancement. (Bhandari and Maurya, 2020) employed the cuckoo search to optimize the plateau limit in sub-histogram equalization.

3. Cuckoo Search

The Cuckoo Search is inspired by brood parasitism of certain cuckoo species. Cuckoos use the nest of another host bird to lay their eggs, thus engage in obligate brood parasitism. The CS algorithm is implemented by taking into consideration this behavior of certain cuckoo species and following three idealized rules (Yang and Suash, 2009):

Each cuckoo has the tendency to lay an egg at a time and deposit it in a randomly picked nest.
The best nests containing high-quality eggs will be delivered over to generation.
There are a limited number of host nests, and the likelihood of the host bird finding the cuckoo egg is $p_a \in [0, 1]$. The host bird will either discard the egg from the nest or leave the nest and create another new one.

From the point of view of implementation, it is assumed that each egg in the nest serves as a solution vector, and a cuckoo can lay only one egg at a time. In that case, no distinction can be made between the cuckoo, egg or nest. All represent the position of the solution in the solution space. By the following equation, the initial solution is obtained.

$$nest_i = Lb_i + rand * (Ub_i - Lb_i) \quad (1)$$

where, $nest_i$ represents the solution vector. Ub_i and Lb_i denote upper and lower search boundaries for a solution, respectively. The position of a new nest (new solution) for dropping the egg is calculated based on the previous best nest position according to the Lévy walk (Yang and Algorithms, 2008). Hence, the global random walk for exploration is performed by a Lévy walk as follows, whereas their steps are determined from a Lévy distribution:

$$nest_i^{(t+1)} = nest_i^{(t)} + \alpha \cdot L(\beta) \cdot (nest_i^{(t)} - nest_{best}^{(t)}) \cdot r \quad (2)$$

where, α denotes the step-size parameter, r represents the uniform random quantity, and $nest_{best}$ stands for best solution vector so far. $L(\beta)$ provides a random walk according to the Lévy flight. The Mantegna's algorithm is an exceptionally efficient and straightforward way of generating the steps of the Lévy flight (Mantegna, 1994; Yang and Algorithms, 2008).

The cuckoo search uses a balanced strategy for a local walk and a global explorative walk, moreover, this is governed by the discovery probability, p_a. A fraction of worse nests discovered by p_a performs the local random walk. The local walk is performed as follows (Yang and Deb, 2014):

$$nest_i^{(t+1)} = nest_i^{(t)} + s \otimes H(p_a - \epsilon) \otimes \left(nest_j^{(t)} - nest_k^{(t)}\right) \quad (3)$$

where $H(u)$ represents a Heaviside step function that randomly selects two different solutions $nest_j^{(t)}$ and $nest_k^{(t)}$. ϵ stands for a uniform random number and s is the step size chosen at random. The symbol $\otimes$ indicates element-wise multiplication. On the other hand, Lévy flights are used to take a random global walk (Yang and Deb, 2014).

4. Material and method

4.1. Problem formulation

Poorly illuminated images do not exhibit the full dynamic range. The pattern of an image histogram can identify the problems with an image, e.g., over or underexposure, brightness, contrast, and dynamic range. In a dark image (underexposed) the pattern of the histogram is centered towards the lower intensity side of the gray scale, while in a bright image the pattern is centered towards the higher intensity side. The gray scale of low- and high-contrast images produces a narrow- and broad-range pattern, respectively. When a poorly illuminated image is enhanced well, the pattern of its histogram covers a wide range and is probably shifted towards the higher intensity side.

Brightness, contrast, saturation, and sharpness are the simplest image control parameters that are generally used in image enhancement. Although these parameters are measured by different techniques, the improper control of one parameter affects the others and all the details of the image. It is often confusing how these parameters are used in image processing to make the image visually pleasing or extract information for machine vision systems. The results of an experimental test are shown separately in Fig. S1. In this test, the brightness of the MATLAB in-built image, namely 'cameraman', 'pout' and 'office' (with increasing exposure) is improved by gamma correction. The MATLAB in-built 'brighten' function is used to shift the intensities of all pixel values in one direction. The brightness is increased when the beta is increased from 0 to 1. By increasing the brightness, the mean intensity of the image increases, as shown by the blue line in each image (see Supplementary, Fig. S1). The corresponding contrast and sharpness for each increase in brightness of the images are shown by red and green lines, respectively. It can be seen that after some improvement in brightness, the slopes of contrast and sharpness become negative, i.e., these parameters decrease as the brightness increases. However, images with low exposure exhibit a positive slopes upto slightly higher level (where the brightness compromises the contrast and sharpness). Hence, it can be said that increases in brightness and contrast oppose each other. After the acquisition, several manipulations of pixels in an image are performed to balance the brightness, contrast, saturation, and sharpness. Additionally, it is necessary to note here that the manipulation of pixels somehow introduces distortion or noise in the image, however, good enhancement manipulations highlight the details in an image and make it visually pleasing. Every image has a different pixel distribution, and this is a subjective matter that causes enhancement empirically. Therefore, the enhancement method should automatically adjust the required contrast and brightness. The Cuckoo Search-based algorithm used in this work evaluates the quality of images defined by the objective function in each iteration by applying the transformation function. After some iteration, the proposed algorithm converges into the best optimal point and yields two images. The two images are fused by a nonlinear weighted function to bring out the balance features of both images. In the following subsection, descriptions of the transformation function, objective function, and method are given.

4.2. Function used

A transformation function is used to enhance images in the spatial domain. It takes each pixel's intensity of the input image to generates the enhanced image. When an image is enhanced automatically using an optimization method, an objective function is utilized to determine the image's quality. Two objective functions, fitness1 and fitness2, are employed in the proposed method.

4.2.1. Transformation function

The transformation of an image in the spatial domain is performed as follow:

$$g(x,y) = T[f(x,y)] \tag{4}$$

where $f(x,y)$ represents the input image, $g(x,y)$ stands for the processed or enhanced image, and T is a transformation operator on f, defined in a neighborhood of (x,y) (Gonzalez et al., 2011). The transformation function proposed by us takes into consideration several factors as follows:

- improving the sharpness or acutance of an image without introducing any subtle details (pixel noise) which appear in smooth or flat regions of the image.
- adjusts the gray-level distribution of the image by stretching the shape of the input–output transformation curve.
- the parameters used in the transformation must specify individual features of the image to enhance the contrast.

In this work, the transformation function that performs both local and global enhancement is modelled as follows:

$$g_s(x,y) = c \cdot f(x,y) - a \cdot m(x,y) + b \cdot \sigma(x,y) \tag{5}$$

$$g_b(x,y) = \left(\frac{g_s(x,y) - L_{index}}{H_{index} - L_{index}}\right)^d \tag{6}$$

where $f(x,y)$ = input image

$m(x,y)$ = local mean image with a neighborhood of 3×3

$\sigma(x,y)$ = local standard deviation image with a neighborhood of 3×3

[L_{index}, H_{index}] = lower and upper limits with a tolerance of 1% that is used for contrast stretching

[a, b, c, d] = the parameters used by the optimization algorithm to produce the desired manipulation in the input image.

The first part of the function is applied to the input image to obtain a sharper and highly detailed image by using local enhancement. Secondly, global enhancement is performed by stretching the shape of the input–output transformation curve. The smoothed image $m(x,y)$ is subtracted from the image $f(x,y)$ to achieve local emphasized details, which are further added to image $f(x,y)$, that give a high boost. The parameter 'a' controls the level of smoothness removal for the purpose of emphasizing. This also averages out the pixel noise which appears in smooth or flat regions of the image. The term $\sigma(x,y)$ represents the local contrast that indicates the degree of variation of pixel values in the neighborhood and by adding this to an image, the edge information (intensity) is highlighted. The parameter 'b' controls the sharpness of edges in the image. The shape of the input–output transformation curve controlled by parameter 'd' facilitates global enhancement. It should be noted that the proposed transformation kernel cannot create additional details, just simply emphasize the details. After performing several experiments, the range of the parameters [a, b, c, d] was set to a $\in$ [0, 1.5], b $\in$ [0, 1.5], c $\in$ [2.0, 2.5] and d $\in$ [0, 1.0]. The Cuckoo Search-based algorithm has to determine the best value of [a, b, c, d] to achieve better image enhancement.

4.2.2. Objective function

The proposed Cuckoo Search-based algorithm uses two types of objective functions to generate two images. These objective functions are defined to adjust the two important image features, contrast and brightness, named as fitness1 and fitness2, respectively.

An image with enhanced contrast contains more edge pixels, sharp edges intensity, and high information content. By improving the intensity of edge pixels and edge numbers, an image becomes sharper. The average information content (entropy) is utilized to measure the details of the image. In the light of this, the function fitness1 ($F1$) is defined as follows (Munteanu and Rosa, 2004):

$$F1(I_e) = \log(\log(SS(I_e))) \times \frac{edgels(I_e)}{M \times N} \times H(I_e) \tag{7}$$

where, I_e denotes the generated enhanced image in each iteration, $M \times N$ represents the size of the image, $SS(I_e)$ denotes the sum of the pixels' intensity of the Sobel sharpness image after applying Sobel Mask to I_e, and $edgels(I_e)$ is the number of edge pixels in I_e. The Sobel edge detection technique is used to generate the edge image of an enhanced image. $H(I_e)$ denotes the average information content of I_e, obtained by the Shannon entropy method.

In general, the brightness of the image is a measure of the mean intensity value of all pixels. The histogram of the image shows its tonal range. It indicates the distribution of pixels over the range of brightness in the image. The image histogram provides the quantity of pixel in the image at each grayscale level:

$$h(r_k) = n_k, \ 0 \le k < L \tag{8}$$

where r_k denotes the kth intensity level and n_k denotes the number of pixels at the intensity level r_k.

Adjusting the brightness of an underexposed image or poorly illuminated image can be modelled by fitness2 ($F2$) as follows:

$$F2(I_e) = \mathrm{D}/std \tag{9}$$

where,

$$D = \frac{1}{M \times N} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) \tag{10}$$

$$std = \sqrt{\frac{1}{L} \sum_{k=0}^{L-1} (h(r_k) - \mu)^2} \text{ where, } \mu = \frac{1}{L} \sum_{k=0}^{L-1} h(r_k) \tag{11}$$

The term std defined in Eq. (9) indicates the variation or dispersion of the number of pixels over the range of brightness in the image. It is noteworthy that the images with enhanced contrast exhibit a broad histogram pattern, i.e., the pattern spread over a wide range of mean values. A high value of std is indicative of an image with high contrast and exhibits a broad histogram pattern. The term D indicates the global mean of pixel intensity. A high value of D increases the brightness of the image and shifts the histogram pattern to a higher brightness level. The ratio of D to std indicates that the global brightness lies within one standard deviation. Improving this ratio ensures that the whole broad histogram pattern is shifted towards a higher brightness level in a controlled manner so that the brightness is enhanced without the loss of details.

4.3. Proposed method

From an implementation standpoint, the nest represents the best set of parameters that has to be optimized in the solution space. Cuckoos (particles swarm) first take a Lévy flight to determine whether the host's nest is suitable for eggs by evaluating some objective criteria of the nest and dropping them into the host's nest. In other words, the particles swarm moves toward the optimal position and explores the search space by a Lévy walk. If the host birds find out the egg is not their own, they leave and make a different nest in a different place. For each solution, a selective crossover is applied in the search space. Formally, the Lévy flight provides varying mutation, and the generation of a new solution based on the biased/selective random walk offers a type of crossover. The selection is accomplished by switching probability p_a, which carries the good solutions, and the new one replaces the worse solutions.

In the proposed method, the Cuckoo Search algorithm is modified to generate two sets of parameters for the transformation function. A cuckoo has to find two nests to host their eggs, and the fitness of the eggs is different from one another. It is assumed that the cuckoos have to deliver two types of eggs, which have different objective values, into two different nests of their hosts suitable for hatching a particular type of egg. Hence a cuckoo finds two nests that have different objective values. It is assumed that a nest holds only one egg, and every cuckoo has to search for two nests. Formally, the number of cuckoos is the number of particles swarms which explore two optimal points by evaluating the corresponding fitness of the solution. The modification can be summarized as follows:

- A cuckoo has to randomly search for two nests for their two eggs.
- The fitness of the two eggs is different.
- The best eggs are dropped into the corresponding best nests and passed down to the next generation.
- A fraction of the worse nests is replaced by new ones.

Following the optimization process, the obtained pair of best nest represents two sets of parameters that are utilized for transformation function to generate a pair of enhanced images - one of high sharpness and contrast, the other with an adjusted brightness. Furthermore, the fusion process fuses this pair of images to balance the contrast and brightness and yield the final enhanced image that is visually pleasing. The whole process can be categorized into two processes: (1) Optimization process and (2) Fusion Process. The optimization process used a Cuckoo Search-based algorithm to obtain two sets of parameters. The pseudocode of the Cuckoo Search-based algorithm is shown in Fig. 1. After the optimization process, a pair of best nests, bestnest1 and bestnest2, are the two sets of parameters which are used to produce a pair of enhanced images. Finally, the pair of enhanced images are fused by the fusion process to generate an enhanced output image with the balance of contrast and brightness.

The fusion process is used to combine the best features of the pair of images. Recently, a new angle of view in the fusion process has been introduced to combine the good qualities of the images so that the deficiencies related to the original images are overcome or reduced (Chen et al., 2010). The fusion process proposed in this paper combines contrast and brightness to improve the perception of information and reduce the deficiencies in poorly illuminated images. In the fusion technique, the weight function is defined as a sigmoid function. First, the absolute averages of the high-contrast image $I1$ and adjusted bright image $I2$ are calculated as follows:

$$A(x,y) = (I1(x,y) + I2(x,y))/2 \tag{12}$$

The weight function $W(x,y)$, used to linearly combine the two images, is defined as a nonlinear function of $A(x,y)$:

$$W(x,y) = 1/(1+\exp(-A(x.y))) \tag{13}$$

The final fused image $R(x,y)$, which combines the characteristics of a high-contrast image and adjusted bright image, is defined as:

$$R(x,y) = W(x,y).I1(x,y) + (1 - W(x,y)).I2(x,y) \tag{14}$$

The flowchart of the proposed method is illustrated in Fig. 2.

5. Experimental results

In this section, the simulation results of standard images and Lathe tool image are described. The standard images are taken from (CVG-UGR Image database, 2018). In this paper, the analyses of seven images, namely a Room, Leopard, Carriage, Cottage, Boat, X-ray, and Lathe tool, have been studied, while the proposed method was applied on 100 standard images from (CVG-UGR Image database, 2018). The selected standard images generally suffer from poor contrast and poor brightness. The grayscale image of the lathe tool was acquired by a Stingray camera (Allied Vision Technologies GmbH) assembled with the combination of Navitar lenses. The experimental setup for capturing the images of the lathe tool is explained in (Mahapatra et al., 2015). In practice, it is possible that a captured image of a lathe tool suffers from poor contrast and brightness due to improper illumination in a manufacturing unit. The results were validated with CLAHE, ESIHE, AGCWD, BEASF (γ = 1.0), CEGAMMA, SECEDCT (γ = 0.5), LIME (lambda = 0.15, sigma = 2.0, gamma = 0.8), BIMEF, DRD, standard PSO-based image enhancement (PSOIE), DE based image enhancement (DEIE), and cuckoo search based image enhancement (CSIE) techniques. The maximum iteration value is the criterion for terminating the optimization process, which is set at 40. The number of cuckoos (swarm) and the switching probability are set at 30 and 0.25, respectively.

5.1. Evaluation of image quality metrics

To evaluate the performance of the proposed method, the assessment criteria are as follows: (1) sharpness index and (2) amount of improvement in contrast and brightness of the original image without introducing unwanted artefacts. The normalized sum of the magnitude of the image gradient is considered to be a simple measure of the sharpness (Gonzalez et al., 2011). To measure the amount of improvement in contrast and brightness, the contrast improvement index (CII) and brightness improvement (BI) were used. CII is a quantitative measure which is defined as:

$$CII = \frac{C_{enhanced}}{C_{original}} \tag{15}$$

where $C_{enhanced}$ and $C_{original}$ denote the contrast of the enhanced and original images, respectively. The contrast C of an image is calculated by using Eq. (7). As CII increases, the quality of the enhanced image improves. The brightness improvement in the output image with regard to the original image is calculated as follows:

$$BI = \frac{m_e - m_o}{m_o} \tag{16}$$

where m_e and m_o denote the mean intensity of the enhanced and original images, respectively.

A full-reference perceptual similarity between the input and enhanced images is quantitatively evaluated by using the gradient magnitude similarity deviation (GMSD). The pixel-wise gradient magnitude similarity (GMS) with a polling strategy can predict the perceptual image quality between the original image and the distorted image (Xue et al., 2014). A low value of GMSD is indicative of less distortion in the input and enhanced images. Ideally, if the output image is not distorted after some manipulation, the GMSD of the input (X) and output (Y) images is expected to be GMSD (X, Y) = 0. However, enhancement generally introduces a GMSD value, but this should be less when the brightness and contrast are improved in a balanced way to ensure a good enhancement method. By considering this fact, for an optimal trade-off between contrast and brightness, the ratio of image enhancement to GMSD is higher. In order to test the relative improvement in contrast and brightness of a proposed algorithm with less distortion, the ratio is calculated as follows:

$$r = \frac{CIR^{\alpha} \times BIR^{\beta}}{GMCD^{\gamma}} \tag{17}$$

Algorithm to determine two sets of parameters

Step 1. Initialization
Initialize the population of cuckoo birds, N
Initialize the switching/ discovery probability, p_a
Initialize the two types of nests randomly

$$nest1_i = Lb_i + rand * (Ub_i - Lb_i)$$
$$nest2_i = Lb_i + rand * (Ub_i - Lb_i)$$

Step 2. Lévy walk
for $i \in \{1, 2, 3, .. N\}$

$$nest1_i^{(t+1)} = nest1_i^{(t)} + \alpha \cdot L(\beta) \cdot (nest1_i^{(t)} - nest1_{best}^{(t)}) \cdot r$$
$$nest2_i^{(t+1)} = nest2_i^{(t)} + \alpha \cdot L(\beta) \cdot (nest2_i^{(t)} - nest2_{best}^{(t)}) \cdot r$$

end for

Step 3. Evaluation
for $i \in \{1, 2, 3, .. N\}$
$F1(I_e)_obj_i \leftarrow F1(I_e)$ where $F1(I_e) \leftarrow f(nest1_i)$, $nest1_i = \{a, b, c, d\}$
$F2(I_e)_obj_i \leftarrow F2(I_e)$ where $F2(I_e) \leftarrow f(nest2_i)$, $nest2_i = \{a, b, c, d\}$
end for
Calculate $[F1(I_e)_obj_{max}, K_1] = \max_{i=1,2,..N}(F1(I_e)_obj_i)$; $bestnest1 = nest1(K_1, :)$
$[F2(I_e)_obj_{max}, K_2] = \max_{i=1,2,..N}(F2(I_e)_obj_i)$; $bestnest2 = nest2(K_2, :)$

Step 4. Store the best: Save the best objective value $F1(I_e)_obj_{max}$ and $F2(I_e)_obj_{max}$
Save the corresponding nest $bestnest1$ and $bestnest2$

Step 5. Abandon a of worst nest
for $i \in \{1, 2, 3, .. N\}$

$$nest1_i^{(t+1)} = nest1_i^{(t)} + s \otimes H(p_a - \epsilon) \otimes \left(nest1_j^{(t)} - nest1_k^{(t)}\right)$$
$$nest2_i^{(t+1)} = nest2_i^{(t)} + s \otimes H(p_a - \epsilon) \otimes \left(nest2_j^{(t)} - nest2_k^{(t)}\right)$$

end for

Step 6. Evaluation
for $i \in \{1, 2, 3, .. N\}$
$F1(I_e)_obj_i \leftarrow F1(I_e)$ where $F1(I_e) \leftarrow f(nest1_i)$, $nest1_i = \{a, b, c, d\}$
$F2(I_e)_obj_i \leftarrow F2(I_e)$ where $F2(I_e) \leftarrow f(nest2_i)$, $nest2_i = \{a, b, c, d\}$
end for
Calculate $[F1(I_e)_obj_{new}, K_{new1}] = \max_{i=1,2,..N}(F1(I_e)_obj_i)$; $bnest1 = nest1(K_{new1}, :)$
$[F2(I_e)_obj_{new}, K_{new2}] = \max_{i=1,2,..N}(F2(I_e)_obj_i)$; $bnest2 = nest2(K_{new2}, :)$

Step 7. Survive
if $F1(I_e)_obj_{new} > F1(I_e)_obj_{max}$
$F1(I_e)_obj_{max} = F1(I_e)_obj_{new}$
$bestnest1 = bnest1$
end if
if $F2(I_e)_obj_{new} > F2(I_e)_obj_{max}$
$F2(I_e)_obj_{max} = F2(I_e)_obj_{new}$
$bestnest2 = bnest2$
end if

Step 8. If the Termination criterion is reached, the process is finished; otherwise, go back to Step 2

Fig. 1. The pseudocode of the Cuckoo Search-based algorithm.

$$CIR = max\left(\frac{C_{enhanced} - C_{original}}{C_{original}}, 0\right) \quad (18)$$

$$BIR = max\left(\frac{m_e - m_o}{m_o}, 0\right) \quad (19)$$

where $\alpha, \beta, \gamma = 1$; *CIR* denotes the contrast improvement ratio - $C_{enhanced}$ and $C_{original}$ are calculated by Eq. (7); *BIR* stands for the brightness improvement ratio - m_o and m_e represent the average intensity of the original and enhanced images, respectively; and *GMSD* measures the perceptual image quality as explained by

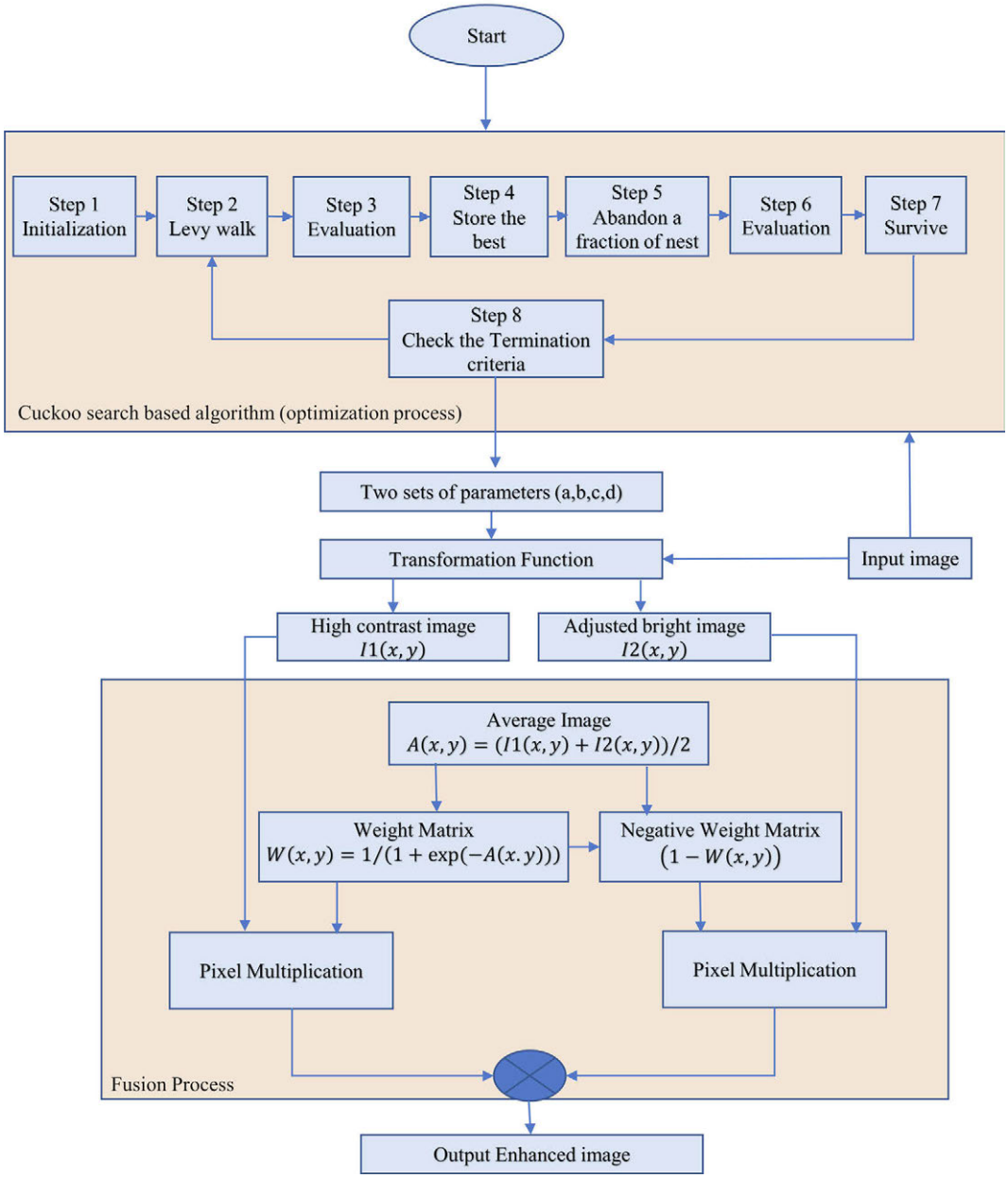


Fig. 2. The flowchart of the proposed method.

(Xue et al., 2014). A value of r equal to zero indicates no improvement in contrast, brightness, or both; a value greater than zero indicates improvement with comparatively less distortion.

5.2. Qualitative and quantitative analyses

The standard images and lathe tool images were applied to the proposed method along with other conventional methods to analyze the improvement in contrast and brightness in a balanced way. The enhanced images of Room and Lathe tools obtained from the proposed method and other enhancement techniques are shown in Figs. 3 and 4. The enhanced results of Leopard, Carriage, Cottage, Boat, and X-ray are separately shown in Figs. S2–S6. Table 1 shows the contrast improvement index (CII), brightness improvement (BI), and sharpness index value of enhanced images for various techniques.

Table 1 reveals that the proposed method has a balanced approach to enhancing the contrast and brightness. Table 2 shows the quality factor (r) of Room, Leopard, Carriage, Cottage, Boat, X-ray, and Lathe tool. Table 2 interpreted that the proposed method improves the brightness and contrast with less distortion, i.e., reduced likelihood of saturation and oversharpness. By evaluating the enhanced images using different techniques, it can be stated that the results are promising, and the proposed method optimizes the contrast and brightness of input images in a balanced way.

By analyzing the Room, Leopard, Carriage, Cottage, Boat, Lathe tool, and X-ray images as well as their enhanced images with different algorithms, it can be noted that results from the proposed method clearly visualize all the details in the images. The output images obtained from the CLAHE method contain more dark and bright areas because the number of pixels was shifted towards both higher and lower values. The BEASF and ESIHE output images were advantageously enhanced, however, these methods not com-

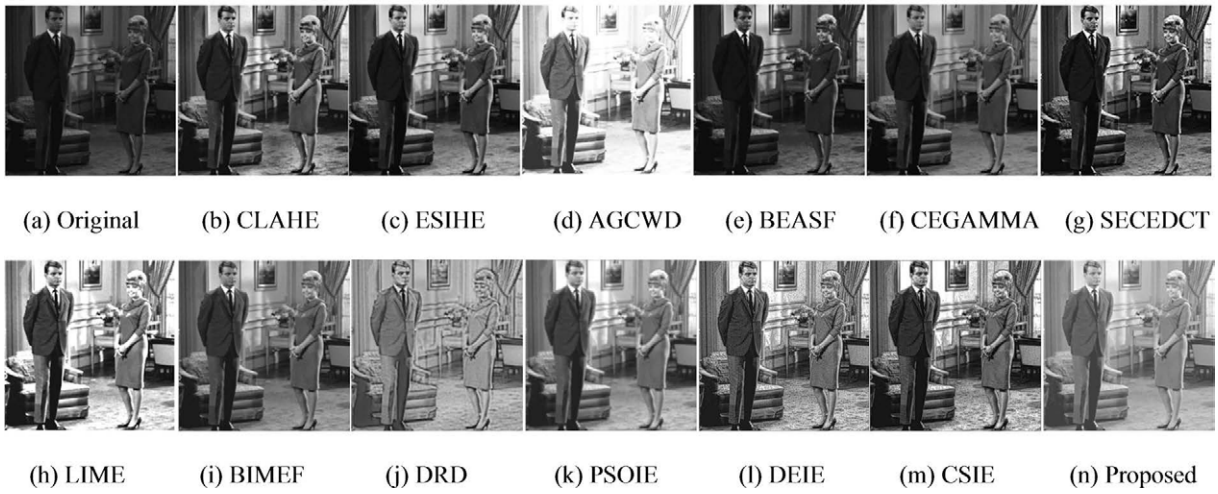


Fig. 3. Enhancement of the image of the Room: (a) Original, (b) CLAHE, (c) ESIHE, (d) AGCWD, (e) BEASF, (f) CEGAMMA, (g) SECEDCT, (h) LIME, (i) BIMEF, (j) DRD, (k) PSOIE, (l) DEIE, (m) CSIE, (n) Proposed Method.

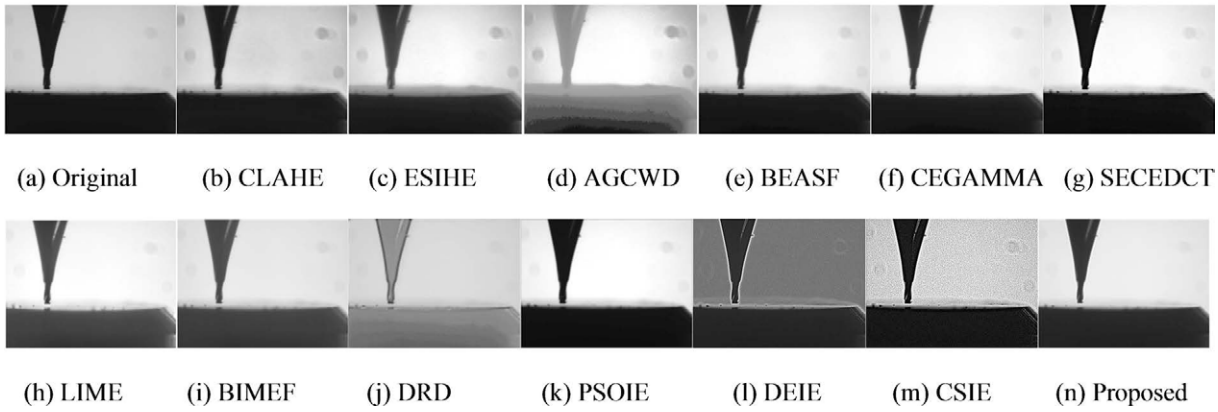


Fig. 4. Enhancement of the image of the Lathe tool: (a) Original, (b) CLAHE, (c) ESIHE, (d) AGCWD, (e) BEASF, (f) CEGAMMA, (g) SECEDCT, (h) LIME, (i) BIMEF, (j) DRD, (k) PSOIE, (l) DEIE, (m) CSIE, (n) Proposed Method.

Table 1
Contrast improvement index (CII), Brightness Improvement (BI), and sharpness of the enhanced images using various enhancement techniques.

Images	Parameters	CLAHE	ESIHE	AGCWD	BEASF	CEGAMMA	SECEDCT	LIME	BIMEF	DRD	PSOIE	DEIE	CSIE	Proposed
Room	CII	1.55	1.27	1.21	0.99	1.28	1.50	1.71	1.39	1.88	1.31	1.62	1.38	1.65
	BI	1.22	1.15	4.29	0.34	0.87	0.89	2.93	1.50	2.77	2.25	2.04	1.85	2.90
	Sharpness	15.23	12.88	15.23	9.84	10.18	21.76	22.32	11.60	21.25	11.55	22.55	20.52	12.63
Leopard	CII	1.21	1.04	1.01	0.97	1.01	1.38	1.28	1.16	1.26	1.06	1.45	1.38	1.30
	BI	0.32	0.26	1.39	0.00	1.15	0.34	1.34	0.37	1.11	0.74	0.51	0.46	0.78
	Sharpness	21.55	17.47	24.08	17.95	19.16	38.73	26.79	18.05	26.86	28.63	40.77	32.48	23.66
Carriage	CII	1.05	1.01	0.88	1.00	0.79	1.16	0.98	1.02	1.00	1.01	1.13	1.16	1.13
	BI	0.75	0.25	1.68	0.17	1.65	0.47	1.48	0.51	1.06	0.86	0.55	0.80	0.94
	Sharpness	26.28	16.84	24.27	17.98	21.78	37.51	30.41	18.37	29.72	19.46	25.24	31.06	23.42
Cottage	CII	1.14	1.07	0.92	1.02	0.87	1.18	1.03	1.03	0.89	1.05	1.12	1.07	1.16
	BI	0.64	0.15	1.54	0.12	1.36	0.38	1.55	0.51	1.09	0.56	0.72	0.73	0.83
	Sharpness	29.67	17.58	27.05	17.32	21.05	39.56	30.28	16.46	27.63	24.68	26.25	33.97	25.20
Boat	CII	1.07	0.98	0.96	0.93	0.89	1.20	0.97	1.01	1.03	1.03	1.25	1.21	1.10
	BI	0.06	0.12	0.22	0.05	0.51	0.13	0.61	0.15	0.32	0.26	0.07	0.15	0.22
	Sharpness	21.71	17.15	20.98	14.08	12.88	31.83	16.77	12.25	15.33	14.77	31.62	27.49	20.35
Lathe tool	CII	1.40	1.53	4.25	1.15	1.14	1.25	1.46	1.38	1.66	1.77	2.50	5.35	2.02
	BI	0.03	0.15	0.33	−1.00	0.25	−1.00	0.39	0.25	0.36	0.20	−0.39	0.00	0.21
	Sharpness	2.96	2.81	3.14	2.38	2.12	4.00	2.32	1.85	2.74	2.80	9.72	18.93	2.74
X-ray	CII	1.30	1.10	1.17	1.03	0.40	1.43	0.84	0.99	0.84	0.99	1.47	1.53	1.31
	BI	−0.05	0.00	0.23	−0.02	0.51	−0.02	0.54	0.20	0.35	0.14	−0.07	−0.09	0.08
	Sharpness	13.67	8.92	10.08	8.36	5.14	15.06	7.38	6.93	7.38	7.93	16.40	23.66	10.04

Table 2
The ratio of improvement to image similarity quality (GMCD), the quality factor (r) of the enhanced images using different enhancement techniques.

Images	CLAHE	ESIHE	AGCWD	BEASF	CEGAMMA	SECEDCT	LIME	BIMEF	DRD	PSOIE	DEIE	CSIE	Proposed
Room	4.90	3.30	3.38	0.00	2.93	3.36	9.47	4.74	9.80	4.82	6.60	3.30	**10.74**
Leopard	0.77	0.32	0.08	0.00	0.06	1.21	2.02	2.17	1.45	0.29	1.38	1.63	**2.28**
Carriage	0.30	0.13	0.00	0.00	0.00	0.69	0.00	0.26	0.01	0.09	0.64	1.03	**1.34**
Cottage	0.74	0.42	0.00	0.07	0.00	0.54	0.24	0.52	0.00	0.19	0.68	0.36	**1.59**
Boat	0.03	0.00	0.00	0.00	0.00	0.24	0.00	0.15	0.06	0.07	0.13	0.20	**0.30**
Lathe tool	0.19	0.74	**4.89**	0.00	0.49	0.00	1.39	1.27	1.77	1.64	0.00	0.00	2.32
X-ray	0.00	0.01	0.28	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.62**

pletely out the details. The study of qualitative measures also shows that the proposed method outperforms both these methods. On the other hand, in some cases the sharpness is higher for CLAHE and ESIHE, which are more suitable techniques than BEASF and CEGAMMA. In most cases, the AGCWD, CEGAMMA, and LIME techniques result in the saturation of brightness. The BI is also higher for these algorithms. However, the result of LIME is comparatively better than those of AGCWD and CEGAMMA. The qualitative analyses also corroborate the subjective evaluation. SECEDCT enhances the contrast more but fails to compensate for the brightness improvement. SECEDCT is better than SECE because it takes into consideration both local and global factors of enhancement. However, in some cases, e.g., in the Leopard image (Fig. S2(j)), pixel noise was introduced in the output image of SECEDCT. DRD and PSOIE enhance well, but in many cases, artefacts like pixel noise, oversharpness, saturation artefacts, halos, etc., were present. It is noteworthy the DEIE and CSIE give a high value of sharpness due to a different transformation function but suffered by the pixel noise. The DRD method introduced flaws in the images because this method is learning-based and depends upon the trained image dataset. Additionally, it was also found in the case of PSOIE, DEIE, and CSIE that a significant degree of variation in the enhancement level was observed when the same image was subject to different runs of the algorithms. In the case of the Lathe-tool image, the AGCWD, CLAHE, DEIE, CSIE, and ESIHE methods yielded an unexpected range of image metrics, and the image quality was visually unsatisfactory. This is not surprising because the histogram equalization-based method is indiscriminate; it may make background noise more noticeable, while diminishing the useful signal. Furthermore, in some cases, the LIME, CEGAMMA and PSOIE methods exhibit a high value of BI but low contrast and sharpness. As a result, these methods over enhanced the input image and reached saturation as an artefact. The local details and sharpness were lost in the image due to saturation. The proposed method reduces the risk of saturation by improving the brightness and contrast in a balanced way. Table 3 shows the average results of 100 standard images from (CVG-UGR Image database, 2018). The paired *t*-test was applied to find the statical significance of the mean between the proposed and conventional methods. The results show that the proposed method is statistically significant (p-value < 0.01) and has a higher mean quality factor (r) when compared to other standard methods. The improvement in contrast is higher and in brightness optimal which shows that the proposed method can prevent problems associated with intensity and saturation as well as provide the optimal trade-off between contrast and brightness. Table 3 also indicates that the AGCWD, CEGAMMA, LIME, and DRD

Table 3
Comparative statistical analysis results of the CVG-UGR image data set for different enhancement techniques.

Algorithms	CII	BI	Quality factor (r)
CLAHE	1.20	0.24	0.50
ESIHE	1.03	0.07	0.16
AGCWD	1.02	**0.96**	0.49
BEASF	1.00	0.04	0.06
CEGAMMA	0.91	0.78	0.31
SECEDCT	1.29	0.16	0.59
LIME	1.07	0.94	1.32
BIMEF	1.05	0.31	0.94
DRD	1.29	0.81	1.06
PSOIE	1.09	0.60	0.89
DEIE	**1.46**	0.34	0.87
CSIE	**1.46**	0.43	1.37
Proposed	1.31	0.59	**1.54**

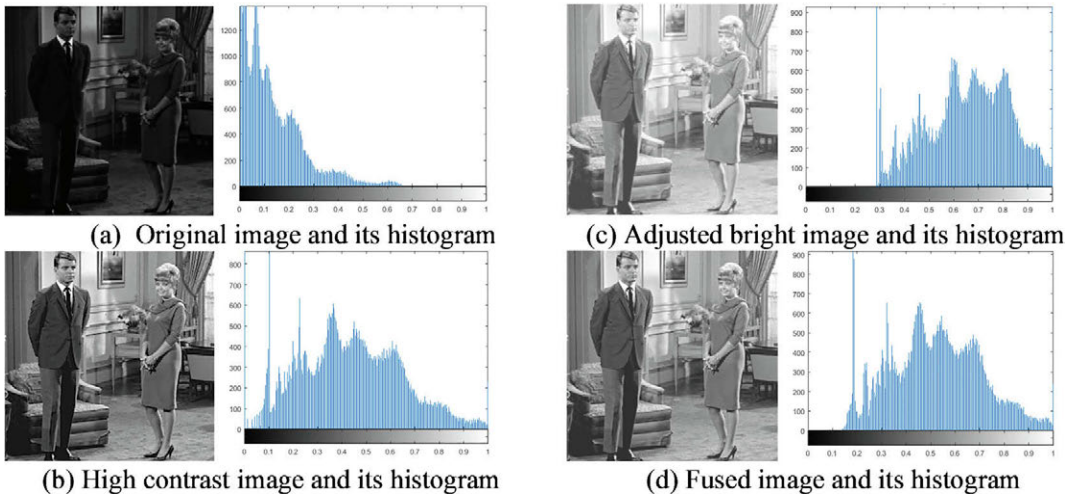


Fig. 5. Simulation results and their respective histograms of the Room image: (a) original, (b) high-contrast, (c) adjusted-bright, and (d) fused image.

Table 4
Mean and standard deviation (STD) of CII and BI of ten different runs of the proposed approach.

Images	CII		BI	
	Mean	STD	Mean	STD
Room	1.6392	0.0136	2.9357	0.0648
Leopard	1.2762	0.0067	0.7865	0.0259
Carriage	1.1409	0.0086	0.9313	0.0206
Cottage	1.1568	0.0054	0.8623	0.0337
Boat	1.1031	9.8968e-04	0.2261	0.0066
Lathe tool	2.0582	0.0289	0.2060	0.0061

methods yield high values of brightness improvement (BI) and subjective evaluation also indicates that these methods introduced the problems associated with the over enhancement and saturation.

5.3. Histogram analyses

The high-contrast, adjusted-bright and the final fused images generated after applying the proposed method on the Room image are shown in Fig. 5. For the high-contrast image (Fig. 5(b)), CII and BI are 1.7005 and 2.2312, respectively. Although the contrast of this image became high, the brightness was not adjusted. For the high-brightness image (Fig. 5(c)), CII and BI are 1.5702 and 4.0190, respectively. Finally, CII and BI for the fused image (Fig. 5 (d)) are 1.6522 and 2.9019, respectively, which show that the proposed method improves the contrast and brightness in a balanced way.

The histogram analyses of the Room image are also shown in Fig. 5. The histogram of the high-contrast image (Fig. 5(b)) is distributed over all levels of intensity. The histogram pattern of the adjusted-bright image (Fig. 5(c)) was shifted towards the high intensity level. The distribution of intensity of the fused image (Fig. 5(d)) was widely spread and slightly shifted as the histogram pattern of adjusted-bright image. Similarly, the distribution of intensity of the fused image using the proposed method was shown a wide and sifted histogram pattern for images of the Leopard, Carriage, Cottage, Boat, and Lathe tool as illustrated separately in Figs. S7–S9. Thus, conclusively, this section explained the quality and characteristics of two generated images according to their name. The high-contrast images yielded high values of CII and their

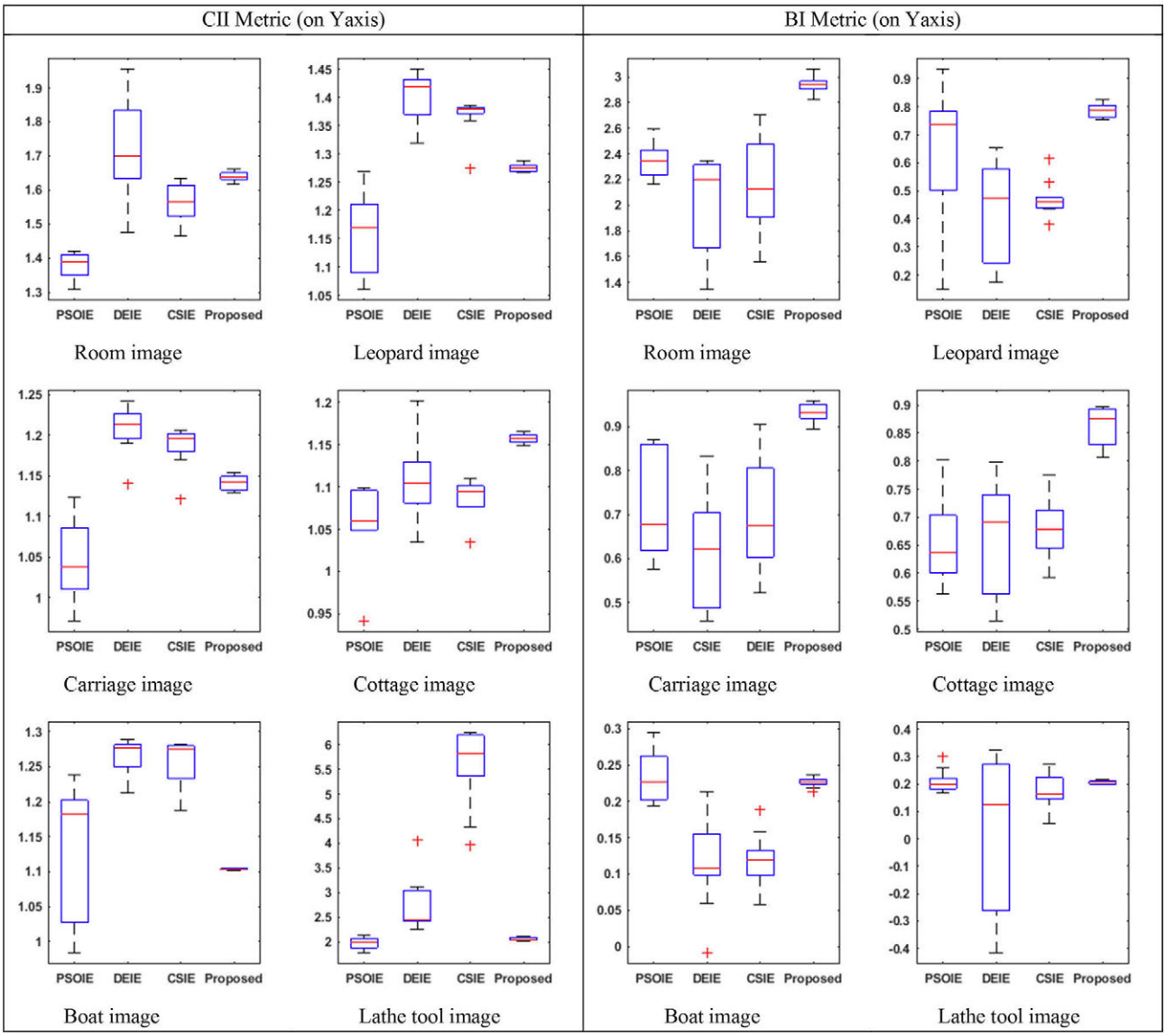


Fig. 6. Box plots of CII and BI for PSOIE, DEIE, CSIE and the proposed method after 10 runs.

histograms were distributed over all levels of intensity. The adjusted-bright images yielded high values of BI and their histograms were shifted towards higher values of intensity. Finally, the histogram distributions of the fused images were widely spread and slightly shifted as the histogram pattern of adjusted-bright image.

5.4. Robustness

To demonstrate repeatability, ten different runs of the proposed approach on each image were performed. The mean and standard deviation of the CII and BI were calculated for the images and tabulated in Table 4. The low standard deviation indicates that different runs produce nearly identical responses.

5.5. Statistical analyses

The box plots for each image were drawn for ten independent runs to analyze the robustness of the proposed Cuckoo Search-based algorithm concerning brightness and contrast, as shown in Fig. 6. The box plot displays the dispersion of CII and BI for PSOIE, DEIE, CSIE, and the proposed method. For proposed method, the box sizes are narrow and converged at the median value. Upon comparison, the proposed method can be regarded as the best choice because the variation is less in both contrast and brightness improvement. CSIE and DEIE have shown a high range of CII due to the oversharpness. Additionally, the median values of BI are high for all the images using the proposed method so the performance of the proposed method is excellent.

6. Discussion and conclusion

In machine vision systems, the problem of poor illumination causes the need for image enhancement. Adjusting the contrast and brightness is difficult in poorly illuminated images. It was determined that by focusing on improving one area of interest does not compensate for another. Furthermore, the existing enhancement algorithms failed to achieve a balance in terms of improvement and caused distortion, e.g., saturation, as well as oversharpness in images. The results of proposed method demonstrated that the contrast and brightness balance in image enhancement. The existing algorithms, namely, CLAHE, ESIHE, AGCWD, BEASF, CEGAMMA, SECEDCT, LIME, BIMEF, and DRD have enhanced the images by improving the relevant features without optimizing the trade-off between the contradictory measures such as brightness and contrast. The advantage of using the Cuckoo Search-optimized image fusion is its excellent utilization of image features by controlling the parameters. The Cuckoo Search algorithm can be easily modified to use one transformation function with two objective functions to generate two images rich in the details of two compatible or contradictory features. Even though the compatible features can be easily combined mathematically to form a single objective function, it is complex to add contradictory measures to a single objective function. A novel way to improve contradictory features in a balanced way is offered by the proposed approach. The comparison of the proposed method with PSOIE, DEIE, and CSIE shown in Fig. 6 demonstrated a well performance and also reveals better convergence and diversity in the optimization of contrast and brightness improvements. The main limitation is that the average computation time of the proposed method is slightly higher than other conventional methods because it uses an iteration based approach to find the optimal balance in contrast and brightness. However, the optimal parameter values are set according to the image automatically so that it achieves better contrast improvement, brightness improvement, and a good ability to prevent the prevention of intensity saturation, yield superior details with no distortion.

The proposed method achieved the intended goal of improving the brightness and contrast in a poorly illuminated image. In the comparison of commonly used image enhancement methods, the proposed approach automatically adjusts the feature parameters according to the image to improve contrast and brightness without introducing any distortion. The results are promising, demonstrating that the proposed method is efficient, robust, and outperforms the conventional techniques in terms of both quality and quantity.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jksuci.2021.07.008.

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